

hw 1

Ian Chi

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1. (a) the sum over all x and y should add up to 1.

$$\sum_{x \in \{0,1,2\}} \sum_{y \in \{0,1\}} c(x+y) = 1 \quad (1)$$

this implies $c = \left(\sum_{x \in \{0,1,2\}} \sum_{y \in \{0,1\}} (x+y) \right)^{-1} c = 1/9$

(b)

$$P(x) = \sum_y c(x+y) = c(x+1) \quad (2)$$

$$P(y) = \sum_x c(x+y) = c(y+3) \quad (3)$$

(c)

$$P(x|y=1) = \sum_x c_2(x+1) \quad (4)$$

where $c_2 = 1/6$

(d) no. the distribution for x changes based on whether y is 0 or 1.

2. (a)

$$\begin{aligned} P_{z_i}(1) &= p \\ P_{z_i}(0) &= 1 - p \end{aligned} \quad (5)$$

(b) the pmf just follows a binomial distribution with $p = p$

$$P(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad (6)$$

$$P(x \geq k) = \sum_k^n \binom{n}{x} p^x (1-p)^{n-x} \quad (7)$$

(c)

$$P(Z_1 = 1 | S_n \geq 1) = \frac{P(S_n \geq 1 | Z_1 = 1)P(Z_1)}{P(S_n \geq 1)} = \frac{1 \cdot p}{\sum_k^n \binom{n}{k} p^k (1-p)^{n-k}} \quad (8)$$

the probability changes since we know one at least is true, each one is more likely to be working.

(d)

$$P(T > m | S_n \geq 1) = \frac{1}{(1-p)^T} P(Z_1 = 1 | S_{n-m} \geq 1) \quad (9)$$

this question is the same as the original problem except the first m failed. the probability that the first m failed is $\frac{1}{(1-p)^T}$ thus we multiply them together and they are independent since no packet affects the outcome of the next packet.

3. (a) the distribution is an exponential distribution but discrete.

$$P(T = k) = (1-p)^{k-1}p \quad (10)$$

(b)

$$1 - \sum_{i=0}^k (1-p)^{i-1}p = (1-p)^k \quad (11)$$

(c) apply

$$\sum k r^{k-1} = \frac{1}{(1-r)^2} \quad (12)$$

for r between 0 and 1. since 1-p is between 0 and 1,

$$\mathbb{E}(T) = \sum k P(T = k) = \frac{1}{p} \quad (13)$$

(d)

$$\begin{aligned} P(T > m+n | T > m) &= \frac{P(T > m | T > m+n) P(T > m+n)}{P(T > m)} \\ &= \frac{1 \cdot (1-p)^{m+n}}{(1-p)^m} = (1-p)^n \end{aligned} \quad (14)$$

it means we sort of can restart our algorithm at the beginning if something fails before.

4. (a) the probability is $\frac{1}{100}$.

- (b) the probability is concentrated into one door and thus the probability is $\frac{99}{100}$.

(c) staying always results in $\frac{1}{100}$. swapping gives probability

$$\frac{99}{100} \frac{1}{99-k} \quad (15)$$

where k is the number of doors opened. the probability is concentrated into $99-k$ doors so the probability is distributed $\frac{1}{99-k}$ ways.

(d) yes. the probability in the equation above is greater than $\frac{1}{100}$ for all $k \geq 1$

5. (a) since each person removes a month from the possibilities of the next persons birthday we see the probability is

$$\frac{12!}{(12-n)!(12)^n} \quad (16)$$

for $n \leq 12$. then 0 for $n > 12$.

(b) the probability is

$$1 - \frac{12!}{(12-n)!(12)^n} \quad (17)$$

. each person removes a $1/12$ probability that the next person has of colliding.

(c) it takes 5 people. $p(\text{same birth month with 5 people}) = .61806$

(d) the probability 3 birthdays collide is $\frac{1}{12}^2$. there should be approx $\binom{n}{3}$ ways fo this to happen, thus

$$P_3(n) = \frac{\binom{n}{3}}{144} \quad (18)$$